

Andreas Ziegler

Robotics & Computer Vision
Researcher/Engineer

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in andreas-ziegler



Summary

I am a broadly trained roboticist with a passion for application-driven robotics, computer vision, and machine learning research. I developed novel algorithms and implemented them to work on real robots. Having lived, studied, and worked in diverse environments—including Zurich, Basel, Lausanne, Tübingen, Paris, Shanghai, and Seoul/Incheon—I appreciate the opportunity to collaborate and exchange ideas with individuals from diverse backgrounds. Diverse perspectives foster creativity and innovation, enabling teams to navigate challenges and deliver impactful results.

I am committed to contributing to a collaborative work environment that embraces experimentation, celebrates learning from failures, and ensures mutual respect and openness. I believe such a culture empowers individuals to take ownership, enabling both personal and collective growth.

As a Postdoctoral Researcher, I seek to grow as an individual contributor while leveraging my leadership experience to foster team collaboration and drive impactful results. By transitioning from individual successes to driving collective achievements, I aspire to stimulate the field of robotics, computer vision, and machine learning.

Personal details

Nationality Swiss

Education

- 2021.06–2026.01 **PhD Candidate**, *University of Tübingen*, Tübingen, Germany
Thesis: Event-based computer vision for fast robot control
○ In collaboration with Sony AI contributing to project ACE.
○ PhD committee (supervisors): Guillermo Gallego, Andreas Geiger, Georg Martius, and Andreas Zell
- 2014.09–2018.04 **MSc ETH in EEIT**, *ETH Zürich*, Zürich, Switzerland
Specialized in: Robotics, Computer Vision and Machine Learning
Master Thesis: A Representation for Exploration that is Robust to State Estimate Drift
○ Examiner: Prof. Dr. Roland Siegwart and Prof. Dr. Davide Scaramuzza
○ Resulted in [13]
Semester Project 2: Map Fusion for Collaborative UAV SLAM
○ Examiner: Prof. Dr. Roland Siegwart and Prof. Dr. Margarita Chli
Semester Project 1: Robust object tracking in 3D by fusing ultra-wideband and vision
○ Examiner: Prof. Dr. Luc Van Gool and Prof. Dr. Otmar Hilliges
- 2009.09–2013.09 **BSc FHO in Electrical Engineering**, *University of Applied Science Eastern Switzerland (HSR)*, Rapperswil, Switzerland
Specialized in: Digital Signal and Image Processing, Embedded Systems and Software Engineering, and Mobile Communication
- 2011.09–2012.08 **Exchange year**, *Shanghai Jiao Tong University*, Shanghai, China
Courses taken: Chinese language, Electrical engineering and Computer Science

Publications

- [1] C. Caccavella, V. Fra, A. Ziegler, G. D'Angelo, and Y. Sandamirskaya, "Efficient semantic understanding from digital foveation," in *Computer Vision – ECCV 2026 Workshops*. Springer Nature Switzerland, 2026. DOI: [arXiv.2609.04088](https://arxiv.org/abs/2609.04088)
- [2] T. Gossard, F. Radovic, A. Ziegler, and A. Zell, "Blurball: Joint ball and motion blur estimation for table tennis ball tracking," in *2026 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, IEEE, Jun. 2026. [Online]. Available: <https://arxiv.org/abs/2509.18387>
- [3] T. Gossard, J. Schmalzl, A. Ziegler, and A. Zell, "Spin detection using racket bounce sounds in table tennis," in *2025 IEEE International Workshop on Sport, Technology and Research (STAR)*, IEEE, Oct. 2025. [Online]. Available: <https://arxiv.org/abs/2409.11760>
- [4] T. Gossard, A. Ziegler, and A. Zell, "Tt3d: Table tennis 3d reconstruction," in *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, IEEE, Jun. 2025, pp. 5821–5831. DOI: 10.1109/cvprw67362.2025.00581 [Online]. Available: <http://dx.doi.org/10.1109/CVPRW67362.2025.00581>
- [5] A. Ziegler, D. Joseph, T. Gossard, E. Moldovan, and A. Zell, "Biasbench: A reproducible benchmark for tuning the biases of event cameras," in *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, IEEE, Jun. 2025, pp. 4994–5003. DOI: 10.1109/cvprw67362.2025.00493 [Online]. Available: <http://dx.doi.org/10.1109/CVPRW67362.2025.00493>
- [6] A. Ziegler, K. Vetter, T. Gossard, J. Tebbe, S. Otte, and A. Zell, "Detection of fast-moving objects with neuromorphic hardware," in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, IEEE, May 2025, pp. 8709–8717. DOI: 10.1109/icra55743.2025.11127960 [Online]. Available: <http://dx.doi.org/10.1109/ICRA55743.2025.11127960>
- [7] T. Gossard, J. Krismer, A. Ziegler, J. Tebbe, and A. Zell, "Table tennis ball spin estimation with an event camera," in *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, IEEE, Jun. 2024, pp. 3347–3356. DOI: 10.1109/cvprw63382.2024.00339 [Online]. Available: <http://dx.doi.org/10.1109/CVPRW63382.2024.00339>
- [8] T. Gossard, A. Ziegler, L. Kolmar, J. Tebbe, and A. Zell, "Ewand: An extrinsic calibration framework for wide baseline frame-based and event-based camera systems," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, IEEE, May 2024, pp. 14534–14540. DOI: 10.1109/icra57147.2024.10610116 [Online]. Available: <http://dx.doi.org/10.1109/ICRA57147.2024.10610116>
- [9] T. Gossard, J. Tebbe, A. Ziegler, and A. Zell, "Spindoe: A ball spin estimation method for table tennis robot," in *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE, Oct. 2023. DOI: 10.1109/iros55552.2023.10342178 [Online]. Available: <http://dx.doi.org/10.1109/IROS55552.2023.10342178>
- [10] A. Ziegler, T. Gossard, K. Vetter, J. Tebbe, and A. Zell, "A multi-modal table tennis robot system," in *RoboLetics: Workshop on Robot Learning in Athletics @CoRL 2023*, 2023. DOI: 10.48550/arXiv.2310.19062 [Online]. Available: <https://arxiv.org/abs/2310.19062>
- [11] A. Ziegler, D. Teigland, J. Tebbe, T. Gossard, and A. Zell, "Real-time event simulation with frame-based cameras," in *2023 IEEE International Conference on Robotics and Automation (ICRA)*, IEEE, May 2023. DOI: 10.1109/icra48891.2023.10160654 [Online]. Available: <http://dx.doi.org/10.1109/ICRA48891.2023.10160654>

- [12] A. N. Horvath et al., “Focus on time: Dynamic imaging reveals stretch-dependent cell relaxation and nuclear deformation,” *Biophysical Journal*, vol. 120, no. 5, pp. 764–772, Mar. 2021, ISSN: 0006-3495. DOI: 10.1016/j.bpj.2021.01.020 [Online]. Available: <http://dx.doi.org/10.1016/j.bpj.2021.01.020>
- [13] T. Cieslewski, A. Ziegler, and D. Scaramuzza, “Exploration without global consistency using local volume consolidation,” in *IFRR International Symposium on Robotics Research (ISRR)*, Hanoi, 2019, IFRR: IEEE, Oct. 2019. [Online]. Available: <https://doi.org/10.5167/uzh-197724>

Work experience

- 2026.09– **Postdoctoral Researcher**, *University of Tronto*, Toronto, Canada, 100%
 ○ Working on event-based state estimation in Tim Barfoot’s ASRL
 ○ Supervision of PhD, MSc and BSc students
 Technologies used: C++, Python, Julia, PyTorch, OpenCV, numpy, Eigen, ROS1/2, git, L^AT_EX
- 2023.11–2024.03 **Research Scientist Intern**, *Sony AI*, Zurich, Switzerland, 100%
 Worked on multi modal camera calibration in project ACE under the supervision of Dr. Raphaela Kreiser and Dr. Naoya Takahashi.
 Technologies used: C++, Python, OpenCV, Ceres, git
- 2021.06–2025.11 **PhD Candidate**, *University of Tübingen*, Tübingen, Germany, 100%
 ○ Working on event-based computer vision for fast robot control in collaboration with Sony AI contributing to project ACE
 ○ Supervision of MSc and BSc students
 ○ Teaching Assistant
 Technologies used: C++, Python, Julia, PyTorch, OpenCV, numpy, Eigen, ROS1/2, git, L^AT_EX
- 2023.11–2024.03 **Research Scientist Intern**, *Sony AI*, Zurich, Switzerland, 100%
 Worked on multi modal camera calibration in project ACE under the supervision of Dr. Raphaela Kreiser and Dr. Naoya Takahashi.
 Technologies used: C++, Python, OpenCV, Ceres, git
- 2022.08–2022.10 **Computer Vision & ML Research Intern**, *Prophesee*, Paris, France, 100%
 Worked on slow motion from frame and event data under the supervision of Dr. Amos Sironi.
 Technologies used: Python, PyTorch, OpenCV, numpy, git, Atlassian tools
- 2018.09–2021.05 **Robotics Engineer**, *MT-Robot AG*, Zwingen, Switzerland, 100%
 Accomplished tasks:
 ○ Development of a computer vision based safety field intrusion detection system
 ○ Improvement of a multi robot collision avoidance system
 ○ Development and maintenance of software for autonomous mobile robots (AMRs), including topics such as multi sensor fusion, mapping, path planning, (multi robot) obstacle avoidance, etc.
 ○ Deputy Scrum Master
 Technologies used: C++, Python, ROS1/2, DDS, OpenCV, CMake, git, Atlassian tools
- 2018.06–2018.09 **Research Assistant**, *University of Zürich, Robotics and Perception Group*, Zurich, Switzerland, 100%
 Continued working on my master thesis project which lead to [13].
- 2018.04–2018.06 **Research Associate Intern**, *Disney Research Zürich*, Zürich, Switzerland, 100%
 Integrated a Leica total station in an existing ROS setup within the PaintCopter project.
 Technologies used: C++, Python, ROS, Ceres, CMake, git

- 2018.02–2018.03 **Research Assistant**, *Laboratory for Orthopaedic Biomechanics at the University and ETH Zürich*, Zürich, Switzerland, 100%
Developed an LED light controller for a microscope setup which contributed to **Horvath2021bj**.
Technologies used: C++, Qt, wxWidgets, CMake, git
- 2017.03–2017.08 **Computer Vision & Robotics Research Intern**, *Pix4D SA*, Lausanne, Switzerland, 100%
Accomplished tasks:
 - Worked on indoor navigation for UAVs
 - Implementation of a filtering method for a robust target detection
 - Investigation of barcode localization and detection algorithms for automatic inventory
Technologies used: C++, ROS, OpenCV, Eigen, Conan, CMake, Jenkins, git
- 2015.08–2018.06 **Software Engineer & System Administrator**, *Accelerom AG*, Zürich, Switzerland, 20%-30%
Technologies used: Java, Groovy, JavaScript, jQuery, CSS, Grails, Hibernate, MySQL, git, Redmine, Tomcat, Apache, SAMBA
- 2014.02–2015.08 **Research Assistant**, *Laboratory for Orthopaedic Biomechanics at the University and ETH Zürich*, Zürich, Switzerland, 100%/20%
Continued my work, provided further consulting and maintenance.
- 2013.11–2014.02 **Research Assistant (Civil service)**, *Computer Assisted Research and Development, University Hospital Balgrist*, Zürich, Switzerland, 100%
Worked on segmentation algorithms for computer-assisted surgical planning.
Technologies used: Matlab, C#, VTK, CVS
- 2013.08–2013.11 **Research Assistant (Civil service)**, *Laboratory for Orthopaedic Biomechanics at the University and ETH Zürich*, Zürich, Switzerland, 100%
Accomplished tasks:
 - Extended and adapted a microscope control software
 - Developed and implemented a stretcher control software
Technologies used: C++, Qt, wxWidgets, CMake, git
- 2008.08–2009.03 **Computer Science Intern**, *ERP sourcing AG*, Wallisellen, Switzerland, 100%
- 2004.08–2008.08 **Electronics Engineer Apprentice**, *Hch. Kündig & Cie. AG*, Rüti ZH, Switzerland, 100%

Independent coursework and training

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|---|---|
| Startup Center
Tübingen | Leadership Talent Academy. Certificate earned on November, 2024 |
| SIY | Search Inside Yourself: Emotional Intelligence for Leadership. Certificate earned on May 28, 2024 |
| Center for
Nonviolent
Communication | NVC Courses and Workshops. Certificates earned in 2024 and 2025 |
| University of
Tübingen | Career Development through Research Funding. Certificate earned on October 7, 2025 |
| University of
Tübingen | Writing successful grant proposals. Certificate earned on July 11, 2025 |
| University of
Tübingen | From Chaos to Structure: Time- and Self-Management for a More Efficient Work Style. Certificate earned on March 9, 2022 |

University of Tübingen	Slidewriting and Storylining. Certificate earned on May 16, 2022
University of Tübingen	Setting and Achieving Goals Efficiently. Certificate earned on May 5, 2022
University of Tübingen	Speedreading. Certificate earned on April 14, 2022
edX	DT-01x: Self-Driving Cars with Duckietown by ETHx on edX. Specialization Certificate earned on Auguts 15, 2021
Coursera	Deep Learning, a 5-course specialization by deeplearning.ai on Coursera. Specialization Certificate earned on March 16, 2018
edX	Autonomous Mobile Robots by ETHx on edX. Certificate earned on April 17, 2014

Awards and Media coverage

- The Guardian AI-powered robot beats elite table tennis players
Schwäbisches Tagblatt Forscherteam der Uni Tübingen entwickelt Tischtennis-Roboter
University of Tübingen Scholarship for the Leadership Talent Academy

Academic services

- Conference workshops Workshop on Neuromorphic Perception for Real World Robotics (NeuRobots), IROS 2025
Workshop on Neuromorphic Robotic Systems (NeuRoSys), RSS 2026
Paper reviews IROS, ICRA, CoRL, CVPR, ECCV, ACCV, RA-L, T-ASE, T-RO

Supervised thesis

- Bachelor thesis Simulating event-based cameras with frame-based cameras, Daniel Teigland
Deep-learning based table tennis ball tracking with an event camera, Genc Ahmeti
Event-camera, camera and robot arm calibration, Levin Kolmar
3D trajectory prediction from event data, Julian John
Pushing an event-simulator towards its limit, Laura Schiller
Adding noise and artifacts to the event-simulator, Steven Krämer
Event-based camera bias optimization, Eric Langlouis
Multi object tracking via event-based motion segmentation with event cameras, Zhiyu Han
Automatic bias optimization for event cameras using Reinforcement Learning, Emil Moldovan
Automatic bias optimization for event cameras using offline Reinforcement Learning, David Joseph
Grasping a table tennis ball using Model Predictive Control, David Grawe
Asynchronous Graph-based Neural Networks for Ball Detection with Event Cameras, Adrian Bytyqi
Spin estimation of a table tennis ball with an event-camera, Maximilian Schnitt
Event-Based vision for a tactile sensor, Alexander Löwe
Ball Detection with event-based asynchronous sparse convolutional networks, Emre Sözbilir
Asynchronous Graph-based Neural Networks for Ball Detection with Event Cameras, Sonja Baumann
Master thesis Spiking neural network for event based ball detection, Karl Vetter
Event-based spin estimation, Julian Krismer
Real-time MPC control of an industrial robot arm for table tennis, Till Köpf
Automatic bias optimization for event cameras using Inverse Reinforcement Learning, Manuel Rauschenberger

Learning-based spin estimation of table tennis ball with an event-camera, Levin Kolmar

Teaching activities

SS 2025 Teaching Assistant: Mobile Robots
AS 2024 Teaching Assistant: Foundations of Robotics
SS 2024 Teaching Assistant: Mobile Robots
AS 2023 Seminar: Robotics and Robot Vision
AS 2022 Teaching Assistant: Introduction to Computer Engineering
SS 2022 Teaching Assistant: Mobile Robots
AS 2021 Teaching Assistant: Deep Learning

Languages

German Mother tongue
English Excellent, Level C1
French Good, Level B1,
Korean Basics, Level A2
Chinese Basics, Level A1

Technical skills

Languages C++, Python, Julia, C, Java
Software packages OpenCV, ROS1/2, PyTorch, Eigen, boost, DDS

Hobbies

Sports Wing Chun Kung Fu, Yoga, Jogging, Mountaineering
Music Drums, Piano, Vocals

Extra-Curricular activities

- Foodsaver at Foodsharing
- Managing a Labdoo hub
- Co-clerk at Switzerland Yearly Meeting (Quakers)
- Member of the Executive Committee at Quaker Council for European Affairs (QCEA)